

DE-LU Day-Ahead Price Forecast: Analysis Report

Forecast market prices from 1 Sep 2026 - 31 Dec 2027, using provided historical data from 1 Jan 2025 - 7 Aug 2026.

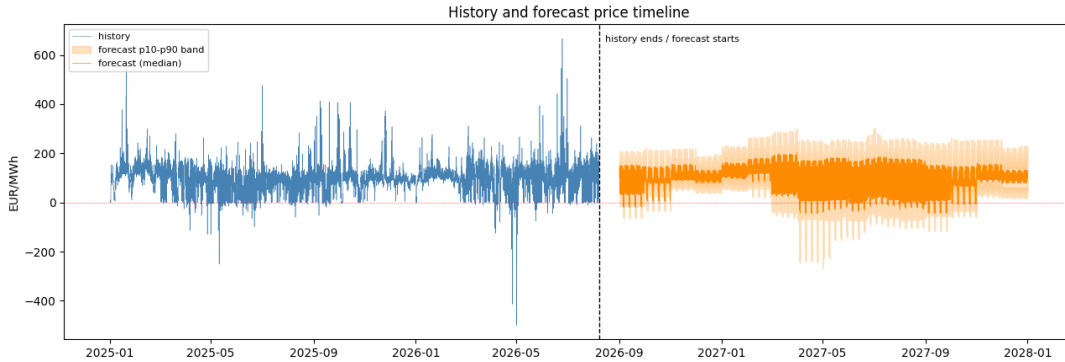


Figure 1: History (actual) and forecast (median + p10–p90 band) on one continuous timeline. Dashed line marks the end of real data.

1 Method

A hybrid two-stage quantile model was used, such that it is able to extrapolate safely the price level. It is composed of a quantile linear regression and a boosted tree model.

- **Linear Model** handles the 4 fundamental features: forecast residual load (load – wind – solar), an implied gas short-run marginal cost (SRMC), an implied coal SRMC, and the forecast renewable share. The model is able to cleanly extrapolate when fuel costs or residual load move outside the training data (historical data). It tries to answer: "What's the overall trend line given today's fuel costs and demand?"
- **Gradient Boosting Regressor (GBR model)** is based on decision trees models and handles the non linear seasonal and time of day shapes. It tries to answer: "Given that we've already accounted for the big trend (linear), how much extra does this particular hour deviate because it's a Tuesday evening in January?"
- Quantiles (p10/p50/p90) fit independently in both stages, summed, then sorted per-row to guarantee $p_{10} \leq p_{50} \leq p_{90}$.
- This model is applied to future data (load, weather, ..) which needs to be created. Due to this, the p10/p90 half-widths are inflated with a factor ($1.0\times$ at the end of history, growing linearly to $1.9\times$ by ~ 18 months ahead) to reflect the extra uncertainty in the *projected* fundamental features themselves, which the backtest (run on real, contemporaneous fundamentals) cannot measure.

2 Fundamental features projection assumptions (2026-09 $\rightarrow$ 2027-12)

We need to project the fundamental features the model uses for the time range that we want to predict the prices. For this, we need to make some assumptions:

- **Load:** climatological profile (mean by month $\times$ hour $\times$ weekend flag, built from history) scaled by an assumed +2.0%/year demand growth. By observing the historical data we can measure a growth of 2.35% (2025 vs 2026 overlap), so we use a growth of 2.0% to be conservative.
- **Solar / onshore / offshore wind:** Using the capacity values of 2025 and 2026, we simply compute the ratio to get the projected 2027 capacity. The historical capacity factor climatology (generation $\div$ installed capacity, by month $\times$ hour $\times$ weekend) is then rescaled by this projected 2027 capacity.
- **Gas (THE day-ahead):** given the restriction of using only the data provided, a simple flat-forward approach is used: compute the most recent 90-day average using the historical data provided and apply a weight given the month. December will be scaled up compared to September, due to the greater demand of gas during winter.
- **EUA (Dec-front):** there was no seasonal pattern so a flat value corresponding to the 90-day average was used.
- **Coal:** no coal price series provided. Different values were used [25%, 40%, 55%] of the gas eur/MWh. The different discounts produce similar prices per month between them, so a value of 40% of the gas EUR/MWh was chosen.
- **Heat rates:** used values from the following table where gas an operating heat rate of 1.77 MWh (thermal)/MWh (electric) ($\approx 56.6\%$ efficiency) and coal 2.27 MWh (thermal)/MWh (electric) ($\approx 44.1\%$).
- **Emission factors:** taken from the following table, where gas emits 0.2028 tCO₂/MWh(thermal) and coal 0.3388 tCO₂/MWh(thermal).
- **Cross-border exchanges:** no data provided; not modelled explicitly. Effects on price are only captured implicitly through the learned residual-load $\leftrightarrow$ price relationship.
- **Quantile-widening schedule** (max +90% width by ~ 18 months): a pure modelling judgment that is not derived from data. It is chosen to reflect a growing uncertainty in the fundamental features projection.

3 Backtest & error metrics

Expanding-window backtest, 6 folds of ~ 2 months each (Sep-2025 $\rightarrow$ Aug-2026), fundamentals taken as known (tests the price $\sim$ fundamentals mapping, not the fundamentals projection itself):

Fold start	Hourly MAE	Hourly RMSE	Monthly baseload MAE
2025-09	15.1	~26	~4
2025-11	21.1	~28	~19
2026-01	14.4	~19	~3
2026-03	14.8	~25	~3
2026-05	21.7	~43	~19
2026-07	15.7	~20	~14
Average	17.1	26.9	10.2

p10–p90 empirical coverage: 91.9% (nominal target 80% – the wider realized coverage is a reasonable margin given the live forecast of the fundamental features). Average bias (median forecast – actual): -1.6 EUR/MWh, close to unbiased. However, the model showed a $+18.8$ EUR/MWh for the fold representing Nov 2025 - Dec 2025 and a -18.4 EUR/MWh for the months of May 2026 - Jun 2026. This clearly represents a limitation of the model during these months, that will be discussed in Section 5.

4 Structural realism checks (final forecast)

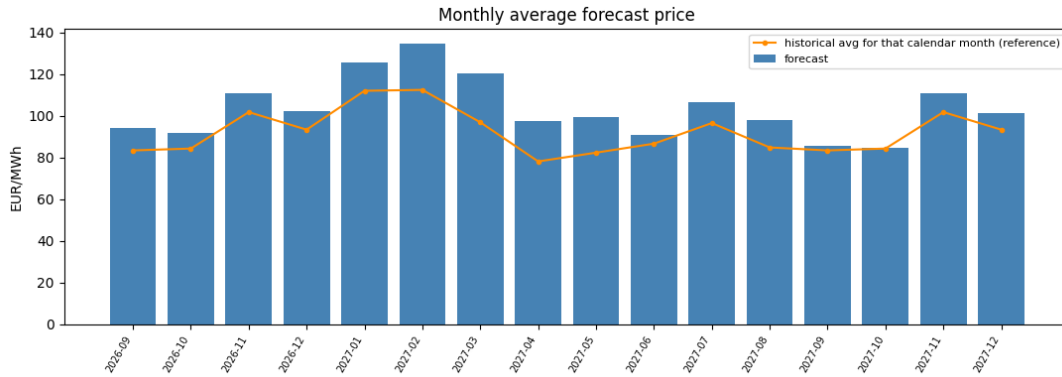


Figure 2: Monthly average forecast price (bars) vs. the historical average for that same calendar month, pooled across all available years (line).

- **Negative-price hours:** 2.8% of forecast hours vs. 6.9% in history. Plausible direction (renewables capacity keeps growing) but likely understated in magnitude: climatology averages away the exceptional high-wind/high-solar tail events that drive the deepest negative-price spells.
- **Seasonality:** as Figure 2 shows, forecast prices peak in Jan/Feb 2027 (~ 129 – 137 EUR/MWh) and trough in Sep/Oct (~ 87 – 90 EUR/MWh), tracking the historical calendar-month reference closely in shape, though consistently somewhat above it – the fingerprint of anchoring gas/EUA on a trailing-90-day level that sits above the full historical average.
- **Consistency with fundamentals:** forecast price tracks residual-load and SRMC-gas with the same sign and rough magnitude of sensitivity as the historical fit (stage-1 linear coefficients all carry the expected sign).

5 Limitations

- No forward curves, weather, or cross-border flow data were available: the entire 2026-2027 fuel and demand trajectory rests on the assumptions in Section 2, by far the largest source of forecast risk (gas alone moved $2.3\times$ min-to-max within the 19 months of observed history).
- Project renewable/load profiles cannot reproduce an atypical weather year, meaning that our model is trained on the weather conditions of 2025 and part of 2026, which won't extrapolate properly to a year where these parameters are very different (climate change)
- No explicit cross-border flow data. Anything price-relevant that depends on how much power Germany and Luxembourg are importing/exporting at a given hour is invisible to the model.
- The coal-price proxy (Section 2) is an invented, unpublished heuristic (the single weakest assumption in the pipeline).
- Backtesting found a distinct issue (unrelated to the fundamental features projections): two folds (Nov-Dec 2025, May-Jun 2026) showed a large, persisting and same direction weekly bias ($+18$ - 20 and -18 - 19 EUR/MWh) even when fed real, observed fundamentals. This points to a genuine, temporary drift in the true fundamental features to price relationship. Basically, these months don't follow the trend seen by the model.

6 With more data

- Forward curves for gas/EUA would be the single biggest accuracy improvement for the far end of the horizon.
- Weather ensembles or multiple historical analogue years for wind/solar (rather than one averaged climatology) would properly widen the renewable-driven tails.
- Real coal price and cross-border flow data would remove two of the weakest current proxies.
- Multi-year capacity build-out plans (grid-operator connection-queue data) would replace the single-ratio capacity extrapolation.